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REGULAR PAPER

A general measure of similarity for categorical sequences

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Abstract Measuring the similarity between categorical sequences is a fundamental process in many data mining applications. A key issue is extracting and making use of significant features hidden behind the chronological and structural dependencies found in these sequences. Almost all existing algorithms designed to perform this task are based on the matching of patterns in chronological order, but such sequences often have similar structural features in chronologically different order. In this paper we propose SCS, a novel, effective and domain-independent method for measuring the similarity between categorical sequences, based on an original pattern matching scheme that makes it possible to capture chronological and non-chronological dependencies. SCS captures significant patterns that represent the natural structure of sequences, and reduces the influence of those which are merely noise. It constitutes an effective approach to measuring the similarity between data in the form of categorical sequences, such as biological sequences, natural language texts, speech recognition

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data, certain types of network transactions, and retail transactions. To show its effectiveness, we have tested SCS extensively on a range of data sets from different application fields, and compared the results with those obtained by various mainstream algorithms. The results obtained show that SCS produces results that are often competitive with domain-specific similarity approaches.

Keywords Categorical sequences · Similarity measure · Chronological order · Matching · Significant patterns

1 Introduction

Categorical sequences are data structured as strings of related or unrelated categories, for which both chronological order and structural features (i.e., subsequences characterizing the intrinsic sequential nature of related sequences) are informatively important. Many types of scientific and business data are in the form of categorical sequences: for instance, biological sequences, natural language texts, retail transactions, etc.

The similarity between categorical sequences is measured through the detection of chronological dependencies and structural features hidden within these sequences. This measure can lead to a better understanding of the nature of these sequences, in the context of their application fields. For instance:

- In biochemistry, each protein has its own unique linear chain made up of 20 possible amino acids, containing structural features, known as conserved domains, which precisely define its biochemical activity. Many different proteins are involved in the same biochemical activities, since they share similar structural features.
- In linguistics, despite the fact that each written work has its own unique sequence of words, structural features that reveal a certain literary style can be pinpointed, making it possible to identify the author, since each author marks his written work with some structural characteristics definitive of his own style.
- In finance, each credit card holder has his own spending behaviour, from which it is possible to extract some sequential factors describing his unique profile. From these sequential factors, it is possible to extract structural features that may predict customers who have a potential risk of bankruptcy.

In the past few years, with the emergence of research areas such as computational biology and text processing, we have seen an increasing need to develop similarity measures that deal efficiently with categorical sequences. The most important known challenges presented by these data, which are only partially addressed by existing methods, are the following:

- It is difficult to extract the information underlying the chronological dependencies of structural features which may have significant meaning.
- Very often, categorical sequences are infected with significant quantities of noise. Unlike numerical sequences, for which we can filter out noise by applying signal processing techniques, categorical sequences require the use of a different, specific set of approaches to handle the non-dependency between the categories making up these data.
- The absence of a measurable similarity relation between the values of the different categories forming these data makes it difficult to measure the similarity between the categorical sequences.
- The high computational cost involved is also an important problem.

- Categorical sequences may include similar structural features with significant meaning in chronologically different positions. This has been ignored by almost all the existing approaches.

The literature reports a number of approaches to measuring the similarity between categorical sequences. One example is the very common Levenshtein distance [19], usually named the "*Edit Distance*", which is calculated by finding the minimum cost required to transform one sequence into another using "*insertion*", "*deletion*" and "*replacement*" operations. Sequence alignment [23] is another commonly used approach that finds the best matching for a pair of categorical sequences by inserting "gaps" in appropriate positions, so that the positions where identical or similar categories occur in the two sequences are aligned.

Both of these approaches have a major drawback due to the fact that they are based on matching of subsequences in chronological order. They break down when applied to sequences comprising similar structural features in chronologically different positions. For instance, protein sequences often have similar conserved domains located in non-equivalent positions when viewed in terms of primary structure, making them difficult to match in chronological order. However, these domains might well be in equivalent positions when viewed in terms of three-dimensional structure [15]. Another drawback the two approaches share is that they yield similarity measures which depend heavily on the costs the user assigns to the "*insertion*", "*deletion*" and "*replacement*" operations in the case of the edit distance, or the "*opening gap*" and "*extension gap*" costs in the case of sequence alignment. This creates ambiguities and complicates the similarity measurement task, especially for sequences of significantly different lengths.

The literature also reports the N-Gram approach [26] for measuring the similarity between categorical sequences. The N-Gram approach is popular for its speed and simplicity. The N-Gram are the set of all possible grams (i.e., patterns) of a fixed length N for which, with an m-letter alphabet, we obtain m^N possible patterns.

It is generally believed that in the N-Gram approach, the restriction to a fixed length N in collecting patterns from the sequences is a major drawback [22]. The value of N is set independently of the intrinsic structure of the sequences, as in the example of the m-letter alphabet, and the length of the sequences. Depending on the value of N, this results in either the collection of patterns representing noise or the exclusion of significant patterns. Moreover, all patterns of length N are collected, without distinguishing between significant and non-significant patterns, which increases the probability of collecting a number of smotifs representing noise.

To the best of our knowledge, the literature does not report any approach that simultaneously addresses all of the challenges cited above. To rectify this shortcoming, in this paper we propose a new general similarity measure named SCS. Our new similarity measure allows us to extract hidden relations between categorical sequences, by capturing structural relations using global information extracted from a large number of sequences rather than merely comparing pairs of sequences. SCS detects and makes use of the significant patterns underlying the chronological dependencies of the structural features, filtering out noise by collecting the significant patterns that best represent the properties of categorical sequences and discarding those patterns that occur by chance and represent only noise. Moreover, SCS measures similarity in a way that more efficiently reflects the structural relationships between categorical sequences, with a worst-case computational cost that is linear with respect to sequence length. In addition, by utilizing an efficient subsequence matching scheme, SCS simultaneously handles the chronological and non-chronological order of the structural features. This allows it to deal with categorical sequences that include similar

structural features with significant meaning in chronologically non-equivalent positions. Our experiments showed that the patterns used in SCS are more significant in terms of representing the natural structural features of categorical sequences and capturing chronological and non-chronological dependencies.

SCS constitutes an effective method for measuring the similarity of categorical sequences. To show this, we have tested it extensively on different data types and compared the results with those obtained by many existing mainstream approaches.

2 The new similarity measure SCS

By applying a new pairwise sequence matching scheme, SCS extracts from a set of categorical sequences a set of patterns with significant meaning, and filters out noise patterns. This is done by examining each pair of sequences for common identical patterns, as well as for patterns that are slightly different, known as "*paronyms*" and "*cognates*". In natural language text, "*paronyms*" such as "*affect*" and "*effect*" are words that are related and derive from the same root, while "*cognates*" such as "*shirt*" and "*skirt*" are words that have a common origin. For a detailed review see [12]. Taking identical patterns, "*paronyms*" and "*cognates*" into account improves the extraction of significant patterns.

After that, the N-Gram algorithm is applied directly on the set of extracted significant patterns, rather than on the original input categorical sequences. The categorical sequences are then mapped onto a new vector space of reduced dimension [10], in which each categorical sequence is represented by a vector. Finally, the measure of the similarity between different sequences is computed simply by calculating the cosine product between the corresponding vectors. This idea is developed in the following sections.

2.1 The main idea of SCS

Very often, in natural language text processing [4], methods such as Latent Semantic Analysis [25] are used to extract hidden relations between documents, by capturing semantic relations using global information extracted from a large number of documents rather than merely comparing pairs of documents. These methods usually make use of a word-document matrix $T(W \times L)$, in which rows correspond to words and columns correspond to documents, where W is the number of possible words and L is the number of documents. The term $T_{i,j}$ represents the occurrence of word i in document j. Although categorical sequences do not contain distinctive patterns like words in natural language text, categorical sequence data analysis is in many respects similar to natural language text analysis. However, the challenge is to be able to identify those patterns that map to a specific meaning in terms of sequence structure and to distinguish significant patterns from patterns resulting from random phenomena.

In much the same way that a word-document matrix is used in natural language text processing to extract the hidden relations between documents, we use a pattern-sequence matrix on the categorical sequences to extract the hidden relations between these sequences. This is done by capturing structural relations using global information extracted from a large number of sequences rather than merely comparing pairs of sequences. Henceforth, we use $T(W \times L)$ to denote the pattern-sequence matrix, in which the term $T_{i,j}$ represents the number of occurrences of pattern i in sequence j, while W is the number of possible patterns, and L is the number of sequences. The significant patterns used to construct T are detected and collected using the matching scheme described in the next subsection.

2.2 The matching scheme

2.2.1 Collection of significant patterns

In this work, a significant pattern is obtained from the matching of a pair of sequences. Let C be a set of categorical sequences, from which X and Y are a pair of sequences. Let x and y be a pair of subsequences belonging respectively to X and Y. Here, the symbol x or y is simply used as a variable, representing any subsequence belonging to the sequence X or Y. A significant pattern is a significantly long matched subsequence of the two sequences X and Y. In this paper, it is any sequence belonging to a matching set $E_{X,Y}$ that is built by collecting all the possible pairs of subsequences x and y that satisfy the following conditions:

$$E_{X,Y} = \left\{ \begin{array}{l|c} x \in X & |x| = |y| \\ y \in Y & |x \cap y| > N_{X,Y} \\ & |x \backslash y| < N_{X,Y} \\ & \forall x', y' \in E_{X,Y} \Rightarrow (x \not\subset x') \vee (y \not\subset y') \end{array} \right\}$$

The symbols x' and y' in the formula are simply used as variables, in the same way as x and y. The expression $(. \not\subset .)$ means that the element to the left of the symbol $\not\subset$ is not included in the one to the right, either in terms of the composition of the patterns or in terms of their positions in their respective sequences. We use the parameter $N_{X,Y}$ to represent the minimum number of matched positions with similar categories between x and y; at the same time, $N_{X,Y}$ is also used to represent the maximum number of matched positions with different categories allowed. A detailed discussion on the choice of $N_{X,Y}$ is provided in the next subsection. Here are a few explanations about the previous formula:

- $|x| = |y|$: means that x and y have the same length.
- $|x \cap y| > N_{X,Y}$: means that x and y include more than $N_{X,Y}$ matched positions with similar categories.
- $|x \backslash y| < N_{X,Y}$: means that x and y include fewer than $N_{X,Y}$ matched positions with different categories.
- $\forall x', y' \in E_{X,Y} \Rightarrow (x \not\subset x') \vee (y \not\subset y')$: means that, for any pair of matched subsequences x' and y' belonging to $E_{X,Y}$, at least one of x and y is not included in x' or y', respectively, either in terms of their compositions or in terms of their respective positions in their corresponding sequences, according to the partial order induced by set inclusion.

By looking for similar patterns in X and Y, the aim of the matching set $E_{X,Y}$ is to capture information shared by X and Y, related to certain chronological dependencies in their structural features. At the same time, by taking into account multiple occurrences of patterns in non-equivalent positions, the matching set $E_{X,Y}$ seeks to capture the structural features in non-chronological order. In fact, with this formula, $E_{X,Y}$ captures pairs of patterns x and y that show a "*within*" chronological similarity, even if they are in non-chronological order from the standpoint of their respective positions within the sequences X and Y.

As an example of the matching scheme, let X, Y, and Z be three categorical sequences for which we want to measure the pairwise similarity, as illustrated in Fig. 1. In this example, by assuming that $N_{X,Y} = N_{X,Z} = N_{Y,Z} = 3$, the matching scheme will match, from each pair of sequences, the pairs of similar patterns with length >3 that also contain a number of mismatches <3 (specified by the matching condition). The pairs of patterns satisfying the matching condition are collected, as shown in Table 1.



Fig. 1 Example showing matching of categorical sequences. *Shaded patterns* correspond to matched significant patterns

Table 1 Pairwise matching scheme Letters in *bold* correspond to mismatches–*shaded patterns* correspond to collected pairs of patterns

Pairwise Matching		X		Y		Z	
		x_1	x_2	y_1	y_2	z_1	z_2
X	x_1	–	–	TM**G**ITC**D** TM**S**ITA**D**	TM**R**ITF**D** TM**S**ITA**D**	TM**GA**TE**D** TM**SI**TA**D**	T**LR**ITAD T**MS**ITAD
	x_2	–	–	TM**G**ITC**D** TM**M**ITE**D**	TM**R**ITF**D** TM**M**ITE**D**	TM**GA**TED TM**MI**TED	T**LR**ITA**D** T**MM**ITE**D**
Y	y_1	TM**S**ITA**D** TM**G**ITC**D**	TM**M**ITE**D** TM**G**ITC**D**	–	–	TMG**A**TE**D** TMG**I**TC**D**	T**LR**ITA**D** T**MG**ITC**D**
	y_2	TM**S**ITA**D** TM**R**ITF**D**	TM**M**ITE**D** TM**R**ITF**D**	–	–	TM**GA**TE**D** TM**RI**TF**D**	T**L**RITA**D** T**M**RITF**D**
Z	z_1	TM**SI**TA**D** TM**GA**TE**D**	TM**MI**TED TM**GA**TED	TMG**I**TC**D** TMG**A**TE**D**	TM**RI**TF**D** TM**GA**TE**D**	–	–
	z_2	TM**S**ITAD T**LR**ITAD	TM**M**ITE**D** T**LR**ITA**D**	TM**G**ITC**D** T**LR**ITA**D**	TM**R**ITF**D** T**L**RITA**D**	–	–

2.2.2 Minimum length of significant patterns

An important feature of our similarity measure is the determination of the length of patterns to be considered significant. It is well known that the longer the patterns, the smaller the chance of their being identical by chance, and vice versa. In fact, according to Karlin's Theorem 1 in [13], the expected length $K_{R,L}$ of the longest common pattern present by chance at least R times out of L m-category sequences $S_1, S_2, \ldots, S_L$, is calculated as follows:

$$K_{R,L} = \frac{\log n(|S_1|,\ldots,|S_L|)+\log\lambda(1-\lambda)+0.577}{-\log\lambda}$$

$$n\left(|S_1|,\ldots,|S_L|\right) = \sum_{1\le i_1\le\cdots\le i_R\le L}\ \prod_{v=1}^{R}|S_{i_v}|$$

$$\lambda = \max_{1\le i_1\le\cdots\le i_R\le L}\left(\sum_{i=1}^{m}\prod_{j=1}^{R}|p_i^{(v_j)}\right)$$

$$\sigma \approx \frac{1.283}{|\log\lambda|}$$

where $p_i^{(v_j)}$ is generally specified as the ith category frequency of the observed vth sequence, while σ is the asymptotic standard deviation of $K_{R,L}$.

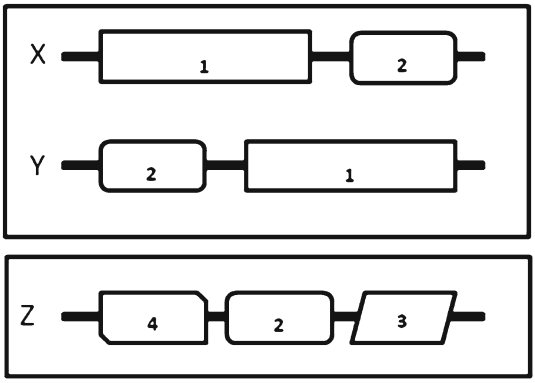


Fig. 2 Application of Karlin's theorem

For a given set of categorical sequences for which we want to detect and extract the significant patterns, we make use of Karlin's theorem on each pair of matched sequences X and Y to estimate one specific and appropriate value for the minimum length of significant patterns $N_{X,Y}$ for the pair. A conventional way of using the theorem would be to estimate, for the whole set of sequences, a single value for the minimum length of significant patterns. Applying Karlin's theorem to each pair of sequences tends to yield a large value for $N_{X,Y}$ and consequently reduces the chance of collecting patterns occurring by chance. Figure 2 gives a typical example to illustrate this idea.

First, let us take a pair of related categorical sequences X and Y, from which we want to extract the significant patterns. Within these sequences, let us suppose that pattern 2 occurs by chance and pattern 1 has a significant meaning in terms of the structures of X and Y. Applying Karlin's theorem, we likely get the longest common pattern present by chance equal to the length of pattern 2. By matching X and Y, this threshold allows us to select pattern 1 as significant and filter out pattern 2. Now, let us consider the case where there is an additional sequence Z that is unrelated to X and Y. In this case, because of the fact that X and Y are less similar to Z, Karlin's theorem will lead to a shorter length threshold, significantly increasing the risk of considering pattern 2 as significant for X and Y. This is why we have chosen to adopt a pairwise application of Karlin's theorem. In this example, pairwise matching of X, Y, and Z highlights the pattern with significant meaning shared by X and Y; at the same time, considering sequence Z within the set of sequences does not disturb the relation between sequences X and Y.

However, due to the fact that Z is unrelated to X and Y, Karlin's theorem will yield a shorter length threshold for the pairs X with Z, and Y with Z, than for the pair X with Y, which increases the risk of considering pattern 2 as significant for the pairs X with Z, and Y with Z. But even if this happens, the similarity between the pair of related sequences X with Y is more significant than the similarity between other unrelated pairs.

By reformulating the theorem proposed by Karlin we can thus say that the expected length $K_{2,2}$ of the longest common pattern present by chance at least 2 times (i.e., $R = 2$) out of 2 m-category sequences (i.e., $L = 2$), X and Y is calculated as follows:

$$K_{2,2} = \frac{\log\left(|X|^2 + |Y|^2\right) + \log\lambda(1-\lambda) + 0.57}{-\log\lambda}$$

$$\lambda = \max\left(\sum_{i=1}^{m}\left(p_i^X\right)^2, \sum_{i=1}^{m}\left(p_i^Y\right)^2\right)$$

Table 2 Sets E_X, E_Y, and E_Z involving sequences X, Y, and Z, respectively

Set	Composition
$E_X = E_{X,Y} \cup E_{X,Z}$	$\{(x_1, y_1), (x_1, y_2), (x_1, z_2), (x_2, y_1), (x_2, y_2), (x_2, z_1)\}$
$E_Y = E_{Y,X} \cup E_{Y,Z}$	$\{(y_1, x_1), (y_1, x_2), (y_1, z_1), (y_2, x_1), (y_2, x_2), (y_2, z_2)\}$
$E_Z = E_{Z,X} \cup E_{Z,Y}$	$\{(z_1, x_2), (z_1, y_1), (z_2, x_1), (z_2, y_2)\}$

where p_i^X and p_i^Y are generally the ith category frequency of the observed sequences X and Y respectively. For each pair of compared sequences X and Y, we use these formulas to calculate the minimum length of matched significant patterns, which is the value to be assigned to $N_{X,Y}$.

According to the conservative criterion proposed by Karlin we can say that, for a pair of categorical sequences X and Y, a pattern observed 2 times is designated statistically significant if it has a length that exceeds $K_{2,2}$ by at least two standard deviations. Thus, in building the matching set $E_{X,Y}$, we extract all the common patterns that satisfy this criterion. This means that, for the pair of sequences X and Y, we calculate a specific and appropriate value of $N_{X,Y} = K_{2,2} + 2\sigma$. In practice, $N_{X,Y}$ is rounded to the largest integer inferior to $K_{2,2} + 2\sigma$. This criterion guarantees that a matched pattern designated as statistically significant (i.e., a pattern that maps to a specific meaning in terms of sequence structure) has less than a 1% probability of occurring by chance.

2.3 Application of the N-Gram algorithm

2.3.1 The pattern-sequence matrix

Let C be a set of categorical sequences. Let X and Y be two different sequences of C, $N_{X,Y}$ the minimum length of the significant patterns, and $E_{X,Y}$ the set of collected pairs of significant patterns. Let E be the set of all possible matching sets, and $N_{\min}$ the minimum value that $N_{X,Y}$ can have, such that:

$$E = \bigcup_{X,Y \subset C} E_{X,Y}$$

$$N_{\min} = \min_{X,Y \subset C} N_{X,Y} + 1$$

To compute an initial pattern-sequence matrix T, we collect and order all the $N_{\min}$ grams from each significant pattern included in E. For a set of sequences made up of m categories, there could be as many as $m^{N_{\min}}$ possible $N_{\min}$ grams, although the number is much smaller in practice. Let E_X be the set of all possible matching sets involving the categorical sequence X (i.e., all the significant patterns that are parts of X), such that:

$$E_X = \bigcup_{Y \subset C} E_{X,Y}$$

In Table 2, we show as an example the composition of E_X, E_Y, and E_Z, the sets of all possible matching sets involving respectively the categorical sequences X, Y, and Z shown as an example in Fig. 1, for which the collected significant patterns are shown in Table 1.

The initial value of the term $T_{i,X}$ is defined as the number of occurrences of the ith $N_{\min}$ gram belonging to the subset E_X. The final matrix T is obtained by removing the rows with

Table 3 Collected $N_{\min}$-grams

Sequences	Patterns	$N_{\min}$ grams	E_X	E_Y	E_Z
X	x_1 TMSITAD	TMSI	3	2	2
		MSIT	3	2	2
		SITA	3	2	2
		ITAD	3	2	2
	x_2 TMMITED	TMMI	3	2	2
		MMIT	3	2	2
		MITE	3	2	2
		ITED	3	2	2
Y	y_1 TMGITCD	TMGI	2	3	2
		MGIT	2	3	2
		GITC	2	3	2
		ITCD	2	3	2
	y_2 TMRITFD	TMRI	2	3	2
		MRIT	2	3	2
		RITF	2	3	2
		ITFD	2	3	2
Z	z_1 TMGATED	TMGA	1	1	2
		MGAT	1	1	2
		GATE	1	1	2
		ATED	1	1	2
	z_2 TLRITAD	TLRI	1	1	2
		LRIT	1	1	2
		RITA	1	1	2
		ITAD	1	1	2

only zero elements. In other words, we discard those $N_{\min}$ grams that are specific to only one sequence. In our experiments, we found that the number of remaining rows W is much smaller than $m^{N_{\min}}$ (i.e., $\ll m^{N_{\min}}$). This property is very important for the next section.

The most important advantage with this new approach is that each sequence in the set of sequences contributes to the capture of the structural features and chronological dependencies of all other sequences in the set. And the more frequently a pattern occurs in the sequences, the more heavily it is represented in the pattern-sequence matrix T. Moreover, the matrix T is filled by using only the grams corresponding to the significant patterns collected, and not all the possible patterns from the original input categorical sequences as in the classical N-Gram approach.

As an example, we build the pattern-sequence matrix of the significant patterns detected in the categorical sequences X, Y, and Z given in Fig. 1, for which the collected significant patterns are shown in Table 1. In this example, the minimum lengths of significant patterns $N_{X,Y} = N_{X,Z} = N_{Y,Z} = 3$, and thus $N_{\min} = 3 + 1 = 4$. Table 3 shows the collected $N_{\min}$ grams from each detected significant pattern from each categorical sequence. In this table, we also show the number of occurrences of the $N_{\min}$ grams in each of the sets E_X, E_Y, and E_Z. After merging the rows corresponding to identical $N_{\min}$ grams, we obtain the pattern-sequence matrix shown in Table 4.

Table 4 The pattern-sequence matrix

		Sequences		
		X	Y	Z
$N_{\min}$ grams	ATED	1	1	2
	GATE	1	1	2
	GITC	2	3	2
	ITAD	4	3	4
	ITCD	2	3	2
	ITED	3	2	2
	ITFD	2	3	2
	LRIT	1	1	2
	MGAT	1	1	2
	MGIT	2	3	2
	MITE	3	2	2
	MMIT	3	2	2
	MRIT	2	3	2
	MSIT	3	2	2
	RITA	1	1	2
	RITF	2	3	2
	SITA	3	2	2
	TLRI	1	1	2
	TMGA	1	1	2
	TMGI	2	3	2
	TMMI	3	2	2
	TMRI	2	3	2
	TMSI	3	2	2

2.3.2 Range of possible values of $N_{\min}$

Because of the risk that the value of $N_{\min}$ may be too small or too large (i.e., problems of feasibility), we evaluated the possible values of $N_{\min}$ for a variety of categorical sequence data sets from different research fields. We used two widely known collections of well-characterized protein sequences, the COG and KOG databases [27], both in their 9 November 2008 versions, comprising 192,987 proteins from unicellular organisms and 112,920 proteins from eukaryotic organisms, respectively. We also used the Reuters-21578 text categorization test collection, one of the most widely used test collections for text categorization research, which comprises 21,578 manually categorized articles that appeared on the Reuters newswire in 1987. We also used the 1,500 categorical sequences generated via a bio-inspired processing from an in-house speech database used in [21], including a number of recorded human voices made up of isolated French letters and numbers. More details about the data sets are provided in the Experiments section. For each data set, we computed, for all possible pairs of categorical sequences X and Y, the length of collected significant patterns $N_{X,Y}$. Table 5 and Fig. 3 present some statistics on the results obtained with the different data sets.

Table 5 shows, for each data set, the number of sequences (Nbr) included, the average length of the sequences (AV length), the value of $N_{\min}$ obtained, and also the average (AV

Table 5 The value of $N_{\min}$ with different categorical sequence data sets

Dataset	Nbr	AV Length	$N_{\min}$	AV $N_{X,Y}$	SD $N_{X,Y}$
COG	192,987	305	3	5.4	3.2×10^{-7}
KOG	112,920	420	3	5.5	5.6×10^{-6}
Reuters	21,578	17,083	3	5.2	3.2×10^{-6}
Speech	1,310	1,480	6	7.9	8.2×10^{-5}

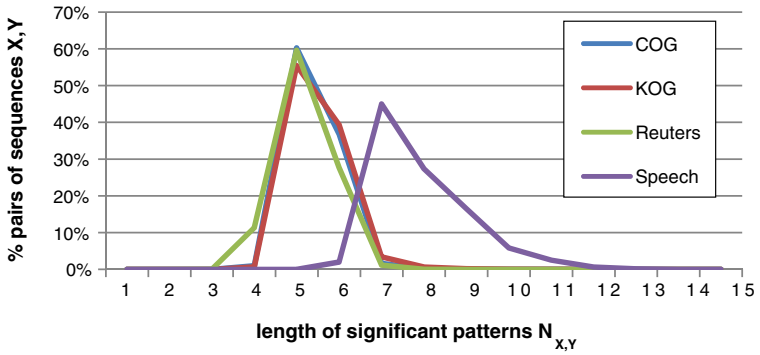


Fig. 3 Percentage of possible pairs of sequences by length of significant patterns

$N_{X,Y}$) and the standard deviation (SD $N_{X,Y}$) of the value of $N_{X,Y}$. In Fig. 3, we also show the percentage of the possible pairs of matched sequences with the same minimum length of significant patterns $N_{X,Y}$.

From Table 5 we see that for all the data sets we obtained $N_{\min}$ values which, in the worst case, allow the building of pattern-matrices with practicable dimensions. The worst case occurs when the pattern-sequence matrix has$m^{N_{\min}}$ rows (i.e., maximum number of possible patterns) with m possible categories. The average value of $N_{X,Y}$(AV $N_{X,Y}$) and its standard deviation SD $N_{X,Y}$, show that the variation in the length of collected significant patterns is rather small. This is confirmed by the results shown in Fig. 3, where we can see that for all data sets, all values obtained for the minimum length of significant patterns $N_{X,Y}$ are comprised within a relatively small restricted range of values.

2.4 The similarity measure

In the pattern-sequence matrix T, each sequence is expressed as a column vector and each pattern as a row vector. This representation is known as a vector space model. Represented in this way, the sequences are seen as points in the multidimensional space spanned by patterns. However, this representation does not recognize related patterns or sequences, and the dimensions are too large [10]. To take advantage of this representation, we perform a singular value decomposition (SVD) on the pattern-sequence matrix T. Let $L = |C|$ and R be the total ranks of T. Thus the matrix T can be decomposed into the product of three matrices, as follows:

$$T = U \times \Sigma \times V^T$$

Table 6 Example of similarity measures

Similarity	Measure
$S_{X,Y}$	0.984
$S_{X,Z}$	0.914
$S_{Y,Z}$	0.929

where U is a $W \times R$ left singular matrix, Σ is an $R \times R$ diagonal matrix of positive singular values, and V is an $L \times R$ right singular matrix. By taking into account only the R' (where $R' \ll R$) largest singular values from the matrix Σ (the choice of R' is discussed in [4]), and their corresponding singular vectors from the matrices U and V, we get the matrix T', the rank R' approximation of T with the smallest error according to the Frobenius norm [11]. Thus, the matrices U, Σ and V are reduced to the $W \times R'$ matrix U', the $R' \times R'$ matrix Σ' and the $L \times R'$ matrix V', respectively, such that:

$$T \approx T' = U' \times \Sigma' \times V'^T$$

Utilizing the singular value decomposition theory [4], the sequences expressed as column vectors in the matrix T' are projected via spectral decomposition onto a new multidimensional space spanned by the column vectors of the matrix V'^T with reduced dimension $R' \ll R$. The only parts of U', Σ', and V'^T that contribute to the value of the ith column of T' are the whole matrix Σ' and $V'^T_{\cdot,i}$ the ith column of the matrix V'^T. Thus, the representation of the sequences in the new R'-dimension space corresponds to the column vectors of the $R' \times L$ matrix $\Sigma' \times V'^T$.

Finally, the similarity measure $S_{X,Y}$ for the pair of sequences X and Y is simply computed by calculating the cosine product of their corresponding column vectors on the $R' \times L$ matrix resulting from the product $\Sigma' \times V'^T$.

The most important advantage of the strategy of transforming the pattern-sequence matrix by spectral decomposition into a new vector representation is that, the similarity measure between different categorical sequences can be computed in the new space using global information extracted from the whole set of sequences rather than merely comparing pairs of sequences. This advantage is made possible by the spectral decomposition that transforms each column-vector in the pattern-sequence matrix into a vector in the new multidimensional space by using the whole set of sequences which gives a global scope to the similarity measure between different vectors.

As an example, we compute the pairwise similarity measures of the categorical sequences X, Y, and Z shown in Fig. 1. First, we perform singular value decomposition of the pattern-sequence matrix shown in Table 4. Next, we compute the similarity measures for each pair of sequences by using the cosine product of the corresponding column vectors of the resulting product matrix $\Sigma \times V^T$ (i.e., we choose $R' = R$). The computed pairwise similarity measures are shown in Table 6.

2.5 Time complexity of SCS

At the stage of collecting the significant patterns, we made use of the fast string matching approach developed by [2], which allows us to find all the locations of any pattern from a sequence X in a sequence Y in time $O\left(|Y|\sqrt{N_{X,Y} \log N_{X,Y}}\right)$. For the singular value decomposition, we utilized the fast, incremental, low-memory and large-matrix SVD algorithm recently developed by [6], which performs the SVD for a R rank matrix $W \times L$ in $O(WLR)$ time with $R \leq \sqrt{\min(W, L)}$.

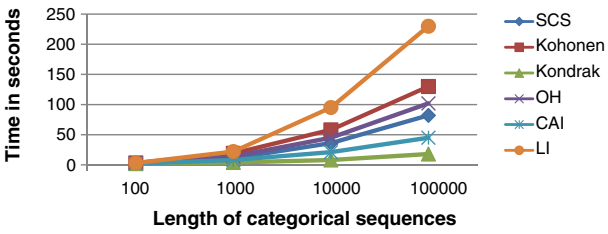


Fig. 4 Execution times obtained with different approaches

3 Experiments

To evaluate the theoretical time complexity of SCS experimentally, and to compare its time efficiency to that of existing approaches, we executed SCS on a selection of four subsets of related categorical sequences. Each of the four subsets includes a large number of related sequences of average length 10^2, 10^3, 10^4, and 10^5, respectively. Then, we compared the execution time obtained with those yielded by a variety of mainstream similarity measure approaches, including those introduced by [7,17,18,20,24]. More details on these approaches are provided in the rest of this section. We report the different execution times for each approach with each subset in Fig. 4.

Figure 4 shows that the execution times obtained by SCS confirm the theoretical time complexity of SCS presented above. From a practical point of view, we see that among the approaches tested, the one developed by [18] obtains the fastest execution time, while that developed by [20] obtains the slowest one. This test shows that efficiency is not the main strength of SCS, even though it obtains a relatively good execution time. However, as we will see in the experiment section, SCS outclasses all of these approaches in terms of effectiveness.

To illustrate the effectiveness of our new similarity measure approach, we have tested SCS extensively on a variety of data sets from different research fields and compared the results with those obtained by several domain-specific mainstream algorithms. In all our experiments, we used these algorithms with their default input parameters. These experiments include tests of SCS on categorical sequences generated from speech data to assess its ability to recognize spoken words and speakers, comparing the results with those obtained by several mainstream algorithms designed to deal with categorical sequences. We also tested SCS more extensively on different protein databases, and compared the results with those of algorithms designed specifically to deal with such data. The aim of the protein data experiments was to illustrate the effectiveness of the general SCS approach in identifying protein sequences according to their functional annotations and biological classifications. Finally, to evaluate its ability to identify related natural language texts, we also tested SCS on the entire Reuters-21578 text categorization test collection, and compared the results with those obtained by algorithms specifically designed to deal with texts and documents.

To evaluate and compare similarity measures, we need an objective quality index. Given that our experiments are performed on data with known classes, it is possible to make use of the well-known ROC Curve approach. Intuitively, a good similarity measure should result in a high similarity value for two sequences belonging to the same class and a low similarity value for those belonging to different classes. For a given sequence X, its class is considered as the positive class and all the other classes together are considered as the negative class. If all the data are classified according to their similarity to X, then an ROC curve can be built. The area under the ROC curve can be used as a quality index of the corresponding similarity measure with respect to the data point X and its class. Obviously, the larger the area under

Table 7 Average length of produced sequences

	M1	M2	M3	M4	M5	F1	F2	F3	F4	F5
"a"	822	923	1245	722	892	1006	1187	2280	1750	1539
"e"	799	1008	883	1921	690	1047	2195	1773	1994	1560
"i"	330	612	578	1361	245	605	1532	1469	1705	804
"o"	335	414	1157	2056	579	503	2925	599	794	749
"u"	512	543	757	1285	447	523	1652	1365	1606	785
"1"	1372	1368	1393	1598	1292	1502	1377	1461	1358	1501
"2"	1201	1020	1252	970	1134	841	1257	994	1227	930
"3"	1306	1118	1378	1216	1274	1306	1413	1115	1404	1227
"4"	1402	1470	1336	1672	1387	1533	1351	1652	1465	1616
"5"	2032	1935	2006	2327	2132	2051	2114	2021	1958	1964
"6"	2036	1991	1974	2293	1950	2227	2173	2255	2030	2211
"7"	1584	1359	1569	1589	1384	1485	1731	1490	1619	1314
"8"	992	999	1200	1206	1089	1050	1156	1102	1128	1177
"9"	1481	1525	1608	1580	1422	1497	1542	1627	1523	1480

the ROC curve is, the greater the discriminative power of the similarity measure approach. Our quality index is defined as the average area under the ROC curve with respect to each data point X. Again, a larger quality index value means greater discriminative power.

3.1 Speech recognition

Speech recognition is the technology that allows computers to automatically identify who says what, by converting the human voice to a type of data much easier to comprehend and analyze using computers. Our aim in making use of these data is to show the effectiveness of SCS on the categorical sequences produced especially for speech recognition. The speech data used in this section come from the in-house speech database used in [21], made up of isolated French letters (i.e., vowels: "*a*", "*e*", "*i*", "*o*", "*u*") and numbers (i.e., "1", . . ., "9") spoken by 5 men and 5 women, with each symbol pronounced 10 times by each speaker. Each recorded speech was used to produce a sequence made up of 20 different events, based on a bio-inspired processing approach [21]. Each pronunciation is thus represented by a categorical sequence made up of 20 different categories. The details about the average lengths of the sequences produced for each letter and number by each speaker are shown in Table 7. The first row contains the list of the different speakers; the symbol "*M*" designates "*male*" and "*F*" designates "*female*". The first column contains the pronounced letters and numbers. The rest of the table contains the average lengths of the sequences produced for each letter and number by each speaker. The produced sequences can be classified either by speakers (i.e., 10 classes) or by words (i.e., 14 classes). In this experiment, we computed the quality index of the results obtained for all classes.

We compared the results obtained using SCS with those yielded by several mainstream approaches. The comparison approaches were the one proposed by [17], based on the set median that has the smallest sum of distances from the other elements; the one proposed by [18], based on the N-Gram approach with a predefined value of N; the one proposed by [24], based on a matching scheme that takes into account the non-chronological order of

Table 8 Quality index with words as classes

	SCS	KOH	KON	OH	CAI	LI
"a"	0.92	0.78	0.78	0.75	0.84	0.74
"e"	0.95	0.74	0.85	0.84	0.82	0.76
"i"	0.97	0.71	0.84	0.75	0.81	0.80
"o"	0.94	0.88	0.88	0.82	0.81	0.74
"u"	0.99	0.79	0.87	0.85	0.78	0.74
"1"	0.97	0.85	0.78	0.80	0.74	0.82
"2"	0.99	0.82	0.90	0.75	0.85	0.73
"3"	0.94	0.83	0.82	0.85	0.82	0.70
"4"	0.92	0.82	0.85	0.75	0.75	0.75
"5"	0.91	0.85	0.76	0.83	0.81	0.80
"6"	0.99	0.79	0.77	0.80	0.85	0.76
"7"	0.97	0.77	0.78	0.84	0.73	0.71
"8"	0.96	0.89	0.80	0.85	0.81	0.82
"9"	0.96	0.79	0.75	0.81	0.78	0.70
Av.	0.96	0.81	0.82	0.81	0.80	0.76

Table 9 Quality index with speakers as classes

	SCS	KOH	KON	OH	CAI	LI
M1	0.93	0.83	0.84	0.80	0.77	0.78
M2	0.90	0.83	0.81	0.75	0.82	0.76
M3	0.95	0.80	0.83	0.76	0.83	0.79
M4	0.98	0.79	0.83	0.73	0.81	0.68
M5	0.92	0.75	0.81	0.77	0.75	0.72
F1	0.95	0.74	0.86	0.78	0.71	0.78
F2	0.96	0.81	0.86	0.77	0.83	0.71
F3	0.96	0.81	0.86	0.77	0.82	0.75
F4	0.98	0.81	0.78	0.76	0.73	0.80
F5	0.95	0.81	0.75	0.81	0.82	0.80
Av.	0.95	0.80	0.82	0.77	0.79	0.76

matched subsequences; the one proposed by [7], based on the longest common subsequences similarity model; and the one proposed by [20], based on sequence alignment.

In Tables 8 and 9 we summarize the results obtained by each algorithm. Each table shows the quality index obtained by each approach (i.e., column) for each subset of sequences belonging to the same class (i.e., row). The last row in each table contains the average quality index obtained by each approach. In Table 8, words are used as known classifications, while in Table 9, speakers are used as known classifications. In Tables 8 and 9 we can see that our approach obtained the best quality indices for both types of classifications, by words as well as by speakers.

We conclude that SCS is able to effectively recognize related categorical sequences generated from the pronounced letters and numbers, whether categorized by pronounced words or speakers, and does so better than the other approaches. We strongly believe that this is because SCS more effectively highlights the significant unseen information behind the

chronological dependencies and structural features within these sequences, thanks to its detection and use of the significant patterns that best represent the natural structure of these sequences, thereby minimizing the influence of those patterns that occur by chance and represent only noise. In addition, the matching technique, which allows us to simultaneously handle the "*within*" chronological order and the "*between*" non-chronological order of the structural features, also plays an important role in reaching these conclusive results.

3.2 Protein sequences

In biochemistry, a protein sequence is a linear chain made up of 20 possible amino acids. Thus, a protein is a categorical sequence made up of 20 possible categories. An important open problem in computational biology is to automatically predict the biochemical activity of a newly sequenced or not yet characterized protein sequence. To achieve this, biologists often compare the non-characterized protein sequence to those that are biochemically well-characterized, and assign to this protein the biochemical activity of the most similar proteins.

In this experiment, we applied SCS to predict the biochemical activities of protein sequences. We tested SCS on a variety of protein data sets and compared the results with those obtained by different mainstream algorithms designed specifically to deal with protein sequences. For instance, we considered SMS, introduced by [16] based on a strict matching scheme that captures the most significant patterns in chronological and non-chronological order; tSMS, introduced by [14], which is an improved version of SMS that allows mismatches; one of the most commonly used Bioinformatics programs, BLAST, introduced by [1] based on the local sequence alignment; the approach introduced by [29] based on short patterns used analogously to the index terms in information retrieval; and the one introduced by [5] based on the cross-entropy measure applied over the collected N-Gram patterns with a fixed value of N. Below, we report the results obtained for the different data sets, with support from the literature and functional annotations.

To demonstrate the effectiveness of SCS in measuring the similarity between protein sequences according to their functional annotations and biological classifications, we have performed extensive tests on the widely known databases COG and KOG [27], and PC (i.e., from the NCBI website). The COG and KOG databases include a classification of proteins encoded in complete genomes. COG and KOG contain 192,987 and 112,920 well-classified protein sequences, respectively. The PC database is a compilation of proteins from the complete genomes of different organisms that have been grouped and manually classified and annotated based on sequence similarity and protein function.

To perform a biological and statistical evaluation of our new similarity measure, we used the three ensembles of randomly generated data sets from [14]: C1 to C6 generated from the COG database, containing respectively 509, 448, 546, 355, 508 and 509 protein sequences; K1 to K6 generated from the KOG database, containing respectively 317, 419, 383, 458, 480 and 388 protein sequences; and finally P1 to P6 generated from the PC database, containing respectively 538, 392, 442, 595, 561 and 427 protein sequences. Each generated subset includes protein sequences with at least 20 biochemical activities, within which each biochemical activity defines a particular class of proteins.

In Tables 10, 11 and 12 we summarize the results obtained by each algorithm on each subset. Each table shows the average quality index obtained by each algorithm (i.e., column) for each subset of protein sequences (i.e., row). The last row in each table contains the global average quality index obtained by each algorithm. The results given in Tables 10, 11 and 12 show that tSMS obtains the best similarity measures over all generated subsets. The results obtained with tSMS are closely followed by those of SCS and SMS, while Wu and Bogan

Table 10 Average quality index on COG

	SCS	tSMS	SMS	BLAST	Wu	Bogan
C1	0.96	0.97	0.93	0.70	0.78	0.84
C2	0.95	0.96	0.95	0.61	0.84	0.88
C3	0.91	0.98	0.95	0.77	0.88	0.82
C4	0.93	0.98	0.89	0.74	0.77	0.82
C5	0.92	0.95	0.93	0.60	0.81	0.84
C6	0.94	0.97	0.95	0.68	0.77	0.86
Av.	0.94	0.97	0.93	0.68	0.81	0.84

Table 11 Average quality index on KOG

	SCS	tSMS	SMS	BLAST	Wu	Bogan
K1	0.91	0.92	0.91	0.65	0.68	0.66
K2	0.91	0.94	0.91	0.55	0.67	0.71
K3	0.92	0.96	0.93	0.58	0.74	0.69
K4	0.86	0.92	0.86	0.54	0.62	0.61
K5	0.88	0.94	0.84	0.70	0.68	0.71
K6	0.88	0.91	0.84	0.75	0.58	0.69
Av.	0.89	0.93	0.88	0.63	0.66	0.68

Table 12 Average quality index on PC

	SCS	tSMS	SMS	BLAST	Wu	Bogan
P1	0.94	0.96	0.93	0.78	0.81	0.76
P2	0.95	0.98	0.92	0.76	0.90	0.79
P3	0.93	0.95	0.94	0.62	0.68	0.83
P4	0.94	0.95	0.91	0.79	0.80	0.80
P5	0.93	0.95	0.92	0.73	0.79	0.78
P6	0.91	0.98	0.94	0.80	0.87	0.93
Av.	0.93	0.96	0.93	0.75	0.81	0.82

obtained less good results. A bit farther behind we find BLAST, which obtains the poorest results.

Although it is not designed especially to handle protein sequences (i.e. it does not take into account the substitution relations between different amino acids), the results yielded by our new approach SCS are very close in quality to the best results obtained by tSMS. Furthermore, the results obtained by SCS are comparable to those of SMS, and much better than those obtained by BLAST, Wu, and Bogan. This performance is especially remarkable if we consider that tSMS and SMS need a substitution matrix as input parameter in order to decide which amino acids should be matched and compute the weights of the significant patterns. In our experiments, the results obtained by tSMS and SMS were made possible by the use of the substitution matrix that maximizes the quality index for each test. This means that one needs prior knowledge about the classes of the protein sequences in order to choose the appropriate matrix for tSMS and SMS. This is precisely the reason for proposing SCS in this paper: SCS is a general measure that does not depend on the use of a substitution matrix.

Table 13 Quality index with Reuters-21578 categories

	SCS	CHIM	WAN	ASLAM
Exchange	0.77	0.83	0.58	0.64
Orgs	0.67	0.82	0.43	0.62
People	0.72	0.75	0.45	0.53
Places	0.75	0.78	0.51	0.58
Topics	0.80	0.85	0.59	0.71
Av.	0.74	0.81	0.51	0.62

3.3 Texts and documents

Measuring the similarity between two texts or documents is a fundamental process in many areas in natural language processing, such as text classification and information retrieval. The key issue is to measure this similarity without explicit knowledge of the statistical nature of these texts. The literature reports a number of approaches developed to measure the similarity between texts and documents. Some of the most recent examples are the one introduced by [8] based on a suffix tree document model, the one introduced by [28] based on the earth mover's distance, and the one introduced by [3] based on an information-theoretic approach. These different approaches have demonstrated their ability to measure the similarity between natural language texts effectively. For this reason, and in the aim of evaluating the performance of our new similarity measure, we decided to perform extensive tests to compare the results obtained by SCS to those of the approaches cited above.

To effectively evaluate the performance of our new approach, we tested SCS on the entire Reuters-21578 text categorization test collection, the most widely used test collection for text categorization research. It comprises 21,578 articles which appeared on the Reuters newswire in 1987. Each article was manually indexed (i.e., classified) according to which categories, from which sets, it belonged to. The category sets (i.e., classes) are as follows: Exchanges (39 classes), Orgs (56 classes), People (267 classes), Places (175 classes) and Topics (135 classes). To make these articles accessible to SCS, they were transformed into categorical sequences by withdrawing spaces and newline marks. This pre-processing concerns only SCS, since the other tested algorithms are designed to handle texts, phrases and words as they are. In this experiment, we computed all quality indices for all Reuters-21578 categories (i.e., classes).

In Table 13 we summarize the results obtained by each algorithm on each of the category sets. The table shows the quality index obtained by each approach (i.e., column) for each subset of articles belonging to the same category (i.e., row). The last row contains the average quality index obtained by each approach. The results summarized in Table 13 show that the approach introduced by [8] obtains the best quality indices over all category sets, followed relatively closely by SCS, while the approaches developed by [3,28] obtain less good results.

In this experiment, despite the fact that the approaches of [3,8,28] and were all designed especially to handle natural language texts by taking into account the semantic concepts underlying words and phrases, and despite the fact that the data used in this experiment were transformed by withdrawing spaces and newline marks to make them accessible to SCS, the results yielded by our new approach are very close in quality to the best obtained results, in comparison with the results obtained by the other approaches.

Table 14 Prediction of biochemical activities of the selected protein sequences

Protein	Organism	Known activity	Predicted activity
Protein sequences with known activities			
AAA24053	Bacteria	β-Galactosidase	β-Galactosidase
AAA69907	Bacteria		
AAA35265	Eukaryota		
AAA23216	Bacteria		
BAA07673	Bacteria		
AAK06078	Bacteria		
AAC48809	Eukaryota	β-Glucuronidase	β-Glucuronidase
AAC74689	Bacteria		
AAA52561	Eukaryota		
AAK07836	Bacteria		
AAA37696	Eukaryota		
AAD01498	Eukaryota		
Protein sequences with unknown activities			
AAV32104	Eukaryota	Unknown	Ribonucleotide-diphosphate reductase
XP_960828	Eukaryota	Unknown	
NP_249831	Bacteria	Unknown	
ACB94306	Bacteria	Unknown	FMRFamide
XP_001675807	Eukaryota	Unknown	
YP_869103	Bacteria	Unknown	
ABK18067	Bacteria	Unknown	ATP-binding cassette
YP_846502	Bacteria	Unknown	
XP_001636168	Eukaryota	Unknown	
ABS67555	Bacteria	Unknown	Neuropeptide precursor
YP_429591	Bacteria	Unknown	
YP_001417212	Bacteria	Unknown	
YP_605034	Bacteria	Unknown	Methyltransferase Type 12
YP_342594	Bacteria	Unknown	
YP_049838	Bacteria	Unknown	

3.4 Application of SCS for the prediction of biochemical activity of proteins

In this section, our new similarity approach is used to predict the biochemical activities of two sets of selected protein sequences from different organisms, obtained from the NCBI website. The first set includes well-characterized proteins, all extensively studied at the biochemical level. This set is used to evaluate the ability of SCS to predict biochemical activities of well-characterized protein sequences. The second set includes not yet characterized proteins whose biochemical activities we sought to predict in this paper, subsequently providing the obtained prediction results to the NCBI staff in the aim of adequately annotating the concerned protein sequences. The database entries and the corresponding organisms for the selected protein sequences are indicated in Table 14.

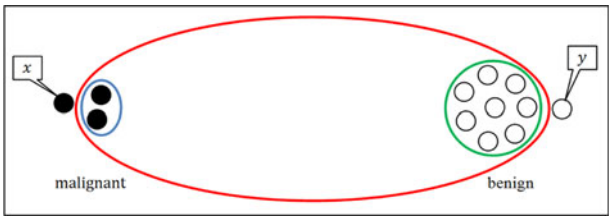


Fig. 5 Difference between *KNN* and *SNN*

To be able to predict the biochemical activities of the target protein sequences, SCS was used to measure the similarity between each of these sequences with all the protein sequences included in the nr database (version of 8 December 2008), the non-redundant protein database maintained by NCBI as a target for BLAST search services, including more than 7 million protein sequences (i.e., very few of which are annotated). Then, an approach called *SNN*, for Significant Nearest Neighbors, inspired by the *KNN* classification approach, was used to systematically select the most significant similar sequences. These were used as the input data set for the alignment-free clustering algorithm CLUSS, developed by [16], given that it recently proved to be more accurate in highlighting the biochemical activities of proteins than the alignment-based algorithms, especially for sequences that are hard to align. A biochemical activity can thus be attributed with high confidence to the uncharacterized protein sequence, if a well-characterized protein within the same cluster is already known. More details about the technique used to select the most similar sequences for each target sequence are provided below.

The *SNN* approach is mainly inspired by the widely known *KNN* classification algorithm [30]. The *KNN* algorithm is a classification method based on statistical theory. It is among the simplest and most widely used pattern classification methods, and is usually employed when there is insufficient prior knowledge about the data distribution. The *KNN* algorithm classifies an object by the vote of its neighbors belonging to the class most common among its K nearest neighbors. For a detailed review of the classification problem, see [9]. The major drawback of *KNN* is that the sensitivity of the results varies depending on the value selected for the parameter K. In particular, difficulties arise when an object from a given class has fewer than K real neighbors from the same class. We can see this clearly in the following example.

The example in Fig. 5 represents the distribution of three objects from the class "black" (i.e., black circles) and nine objects from class "white" (i.e., white circles) according to their pairwise closeness. In Fig. 5, it is clear that black and white circles are well separated. However, *KNN* is not able to distinguish which candidates are actual neighbors. As a clear example of this, if we select $K = 10$, objects within the region bounded by the red line are considered to be neighbors of x as well as of y. This assigns to x and y the same classification, which is clearly not the case. This directly increases the rate of false positives and false negatives in the classification process.

To deal with these drawbacks, here we present *SNN* that dynamically adjusts the value of K in the *KNN* approach. One of the major advantages of *SNN* compared to *KNN* is that *SNN* is able to detect the genuine nearest neighbors instead of choosing a fixed number of neighbors that may poorly reflect the data distribution. This has a direct impact on classification precision, which is explained in the following example.

In Fig. 5, *SNN* is able to distinguish which objects are really neighbors. Objects within the region bounded by the blue line are considered neighbors of x, while objects within the region bounded by the green line are considered neighbors of y. This has the advantage of classifying the object x as 100% in class "black" and y as 100% in class "white". One of the

major advantages of *SNN* compared to *KNN* is thus that *SNN* discriminates more accurately between neighborhoods of different sizes.

The *SNN* approach makes use of a systematic method for deciding which object in a given set of objects to retain as most similar to a target object. We first separate all the objects belonging to this set into two groups, one being the group of candidates highly similar to the target object, and the other of low similarity. This is done by sorting all objects in decreasing order of similarity and computing a separation threshold according to the maximum inter-class inertia method, based on the Koenig-Huygens theorem, which gives the relationship between the total inertia and the inertia of each group relative to the center of gravity. In our case we have just two groups, the high similarity group and the low similarity group. The procedure is described as follows:

Let R be the uncharacterized protein sequence to be predicted, and let F be the set of obtained similarity measures between the sequence R and all the sequences from the nr database, with F_L the subset of low similarity measures, and F_H the subset of high similarity measures, such that:

$$F_L \cup F_H = F$$
$$F_L \cap F_H = \emptyset$$
$$\forall X, Y \in F | X \in F_L, Y \in F_H \Rightarrow S_{R,X} < S_{R,Y}$$

where $S_{R,X}$ and $S_{R,Y}$ are the similarity measures obtained between sequences R and X, and R and Y, respectively. The symbols F_L and F_H are simply used as variables representing all possible separations of F according to previous equations. By making use of the Koenig-Huygens theorem, the total inertia I is calculated as follows:

$$I = \sum_{i \in F_L} \left(S_{R,i} - \bar{S}_{F_L}\right)^2 + \sum_{j \in F_H} \left(S_{R,j} - \bar{S}_{F_H}\right)^2 + \left(\bar{S}_{F_L} - \bar{S}_{F_H}\right)^2$$

where $S_{R,i}$ and $S_{R,j}$ the obtained similarity measures of sequences R and i, and R and j, such that i and j belong to the subsets F_L and F_H, all respectively; and $\bar{S}_{F_L}$ and $\bar{S}_{F_H}$ are the means (i.e., centers of gravity) of subsets F_L and F_H, respectively. The best separation of F is the subsets F_L and F_H that maximize the value of the total inertia I in the previous equation. Then, the most significant similar sequences to be used as input data for the clustering process are the subset of protein sequences corresponding to the subset F_H maximizing I the total inertia.

Table 14 shows the predicted biochemical activities of the target protein sequences. For the set of well-characterized sequences, the clustering has predicted exactly the appropriate biochemical cluster for each protein. For the set of target protein sequences with unknown biochemical activities, the clustering has put each uncharacterized sequence in a cluster containing an already well-characterized protein. Consequently, the activity of the well-characterized protein is assigned to the uncharacterized protein sequence (See Table 14).

4 Discussion

The excellent results obtained in this paper on different types of categorical sequences from different application fields clearly show the effectiveness of our new general method and its advantage over existing domain-specific mainstream methods for measuring the similarity between categorical sequences. First, the results obtained with speech data show that

SCS measures the similarity between pronounced letters and numbers more effectively than other approaches designed to perform the same task on categorical sequences. Second, the results obtained with the protein sequences show that, despite the fact that SCS does not take into account the substitution relations between different amino acids, it is competitive with approaches designed especially to deal with protein sequences. Third, the results obtained with natural language texts show that, even though the data used in this experiment were handled by SCS blindly by withdrawing spaces and newline marks, SCS was able to highlight the related texts as well as the approaches designed to deal with these data by taking into account the semantic relations between words and phrases.

In conclusion, SCS effectively highlights the significant unseen information behind the chronological dependencies and structural features within different types of categorical sequences from different application fields. This is possible because it detects and uses the significant patterns that best represent the natural structures of these sequences, and minimizes the influence of those patterns that occur by chance and represent only noise. In addition, the matching technique, which allows us to simultaneously handle the "*within*" chronological order and the "*between*" non-chronological order of the structural features, plays an important role in reaching these conclusive results.

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